

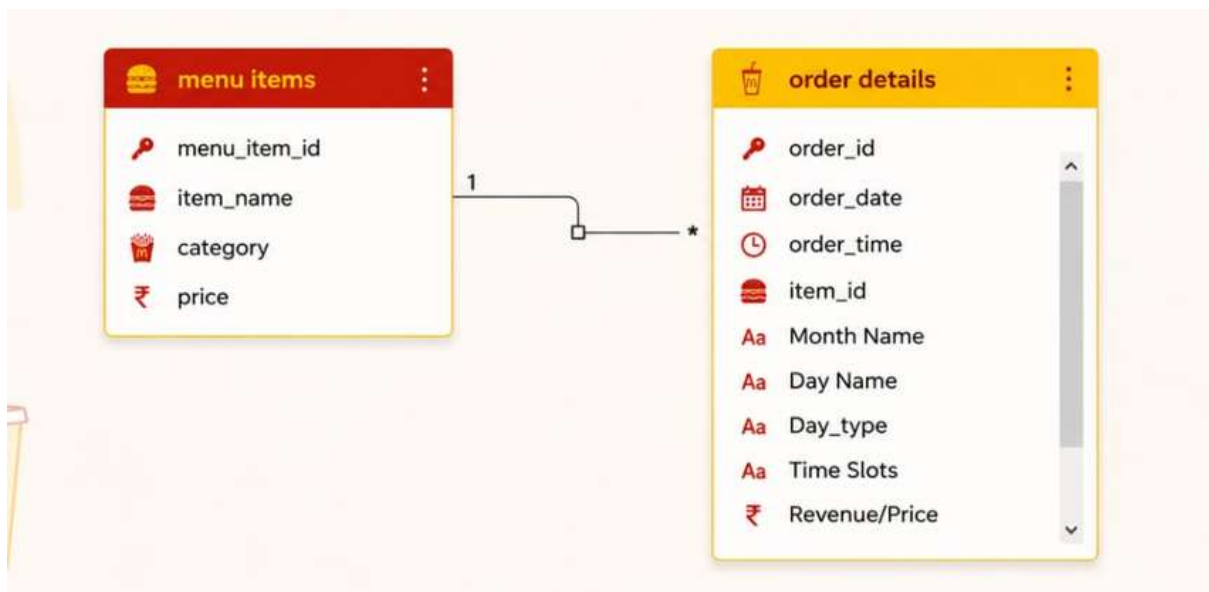
EDA REPORT

Exploratory Data Analysis (EDA) Report

McDonald's Sales & Order Pattern Analysis

1. Introduction

The purpose of this Exploratory Data Analysis (EDA) is to understand transactional ordering behavior and sales performance of McDonald's using order-level and product-level datasets.



2. Objective

The analysis focuses on:

- Sales trends
- Revenue generation

- Product performance
 - Customer ordering behavior
 - Time-based ordering patterns
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3. Dataset Summary

Table 1: Order Details

Feature	Description
Total Rows	12,234
Columns	4

Columns:

- Order ID
 - Order Date
 - Order Time
 - Item ID
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Table 2: Menu Items

Feature	Description
Total Rows	32

Feature	Description
Columns	4

Columns:

- Menu Item ID
 - Item Name
 - Category
 - Price
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4. Initial Data Understanding

Observations

- Each row in the order details table represents a single ordered item.
 - Multiple rows may belong to the same order ID.
 - Menu table acts as the master lookup table.
 - Revenue calculation requires joining both tables.
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5. Assumptions

- Each Row Represents One Quantity Sold
- No Product Returns or Cancellations
- No Store-Level Differentiation
- Dataset Is Considered Complete

6. Data Quality Checks

Missing Values

No major missing values observed in:

Duplicate Records

Checked for:

- Duplicate Order IDs with same Item IDs
- Duplicate menu items

No critical duplication found.

7. Data Transformation Requirements

Since the dataset lacks analytical attributes, additional columns need to be created.

Derived Columns Required

Column	Purpose
Month	Monthly trend analysis
Day Name	Weekday/weekend analysis
Hour	Peak hour analysis
Time Slot	Morning/Evening segmentation

Column	Purpose
Weekend Flag	Behavioral comparison

8. Revenue Analysis

Revenue Calculation

Revenue is calculated as:

Revenue = Count of Ordered Item × Item Price

- Every row is treated as quantity = 1.
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9. Sales Trend Analysis

- High-performance periods
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10. Product Performance Analysis

Most Sold Products

Products with highest transaction count indicate:

- Customer preference
 - High-demand items
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Least Sold Products

Low-performing products may indicate:

- Low customer interest
 - Poor pricing strategy
 - Weak visibility on menu
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9. Category Performance Analysis

Revenue contribution by category helps determine:

- Strongest product categories
 - Most profitable menu sections
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10. Order Pattern Analysis

Peak Order Hours

Important for:

- Staff allocation
 - Kitchen optimization
 - Delivery planning
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Peak Order Days

Analysis helps identify:

- Weekend traffic

- Daily business patterns
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11. Time Slot Analysis

A custom time-slot classification can be created:

Time Range	Slot
6 AM – 12 PM	Morning
12 PM – 3 PM	Noon
3 PM – 7 PM	Evening
7 PM – 12 AM	Night

This improves behavioral analysis.

12. Key Insights

Sales Insights

- A small number of products likely contribute to most revenue.
- Peak revenue periods are expected during evenings and weekends.
- Certain categories dominate overall sales.

Customer Behavior Insights

- Customers tend to order during meal timings.
- Repeat product ordering patterns may exist.
- Fast-food demand is highly time-dependent.

Operational Insights

- Staffing should be increased during peak hours.
 - Inventory planning should prioritize top-selling products.
 - Low-selling items may require strategic review.
 - Run promotions during low-sales periods
 - Rebranding ,Discounting, Removal from menu
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13. Limitations

Dataset Limitations

No Customer Data

Cannot analyze:

- Customer loyalty
- Repeat customers
- Demographics

No Quantity Column

Assumption:

- Every row = quantity 1

This may slightly affect revenue precision.

No Store Location

Cannot perform:

- Regional analysis
- Branch performance comparison

No Cost/Profit Data

Cannot calculate:

- Profit margins
 - Net profit
 - Operational profitability
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14. Dashboard Design & Visualization Strategy

Dashboard Objectives

The final stage of this project focuses on transforming analytical insights into interactive business dashboards. The dashboards are designed to provide management-level visibility into sales performance, customer ordering behavior, and operational trends.

The dashboards are designed to answer the following business questions:

Sales Dashboard

- total revenue generated
- categories contribute the most revenue
- top-selling products
- sales changing monthly
- periods generate maximum revenue



Order Pattern Dashboard

- Time during which most orders are placed
- Most busy days
- What are the peak operational hours?



15. Conclusion

The EDA successfully identifies:

- Product demand patterns
- Revenue trends
- Time-based ordering behavior
- Operational peak periods

The project provides a strong foundation for:

- Dashboard creation
- Business intelligence reporting
- Data-driven decision-making

The integration of Power Query, Power Pivot, and dashboards transforms raw transaction data into meaningful business

insights capable of improving operational efficiency and sales strategy.
